Convergent evolution of burrowing genes in wombats revealed by comparative genomics

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Abstract

Burrowing has evolved repeatedly in marsupials, yet the genetic basis of digging behaviour is poorly understood. We compared 42 marsupial genomes and tested 1,204 genes for episodic positive selection on burrowing lineages. Three genes (BRW1, DIG2 and CLAW3) showed convergent signatures of selection, and site-level models identified 25 codons under diversifying selection. Our results suggest that a small set of developmental genes underlies repeated transitions to a fossorial lifestyle.

Introduction

Fossorial mammals share skeletal and behavioural adaptations for digging. Quokkason et al. (2015) described burrowing-associated expression changes in wombat forelimbs, and Numbat and colleagues later proposed that claw morphology evolves under strong selection. Whether the same genes are repeatedly targeted by selection across independent burrowing lineages remains an open question that comparative genomics can address.

Here we assemble a dataset of 42 genomes spanning four independent origins of burrowing and apply branch-site models of codon evolution to test for convergent episodic selection.

Materials and methods

All sampling procedures were approved by the University of Wombatia ethics board (protocol UW-17-42). Coding sequences were aligned with a codon-aware aligner and a maximum-likelihood phylogenetic tree was inferred under a GTR+G model. Burrowing lineages were labelled as foreground branches.

Gene-wide episodic selection was tested with a branch-site random-effects model, and individual codons were tested with a mixed-effects model of episodic diversification. P values were corrected for multiple testing with the Holm procedure.

Results

Of 1,204 genes tested, 31 showed evidence of episodic positive selection on at least one burrowing lineage (corrected $p < 0.05$). Three genes were selected on all four burrowing lineages (Table 1).

Table 1. Genes under selection on burrowing lineages. BRW1: 12 sites; DIG2: 9 sites; CLAW3: 4 sites.

Figure 1. Phylogeny of the 42 marsupials; burrowing lineages are shown in bold.

Discussion

Convergent selection on BRW1, DIG2 and CLAW3 suggests a shared developmental route to fossoriality. In 2021 we resequenced two additional wombat populations and recovered the same selected sites, although these data were not part of the analyses reported here.

Data availability

All code is available at https://github.com/quokkalab/marsupial-pipeline and the complete analysis history at https://usegalaxy.org/u/aquokka/h/marsupial-run. Raw reads are archived at Zenodo (https://doi.org/10.5281/zenodo.1234567).

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